[Alkemy]
Designed by humans. Powered by AI.

LUISS

AI Productivity analisys.

Beyond which AI usage threshold does rework destroy profit margin?

Presented by :

Group SEA:
· Simone Cavallaro (815441)
· Andrea Mammano (821311)
· Elisa Presciutti (812891)

Alkemy · Machine Learning Project
· LUISS A.Y. 2025/2026

[Alkemy]
Designed by humans. Powered by AI.

LUISS

Beyond which AI usage threshold does rework destroy profit margin?

AI makes tasks faster.

But faster does not always mean more profitable.

We asked: where does the gain go?

Is there a usage level beyond which **AI** starts **costing more than it saves**?

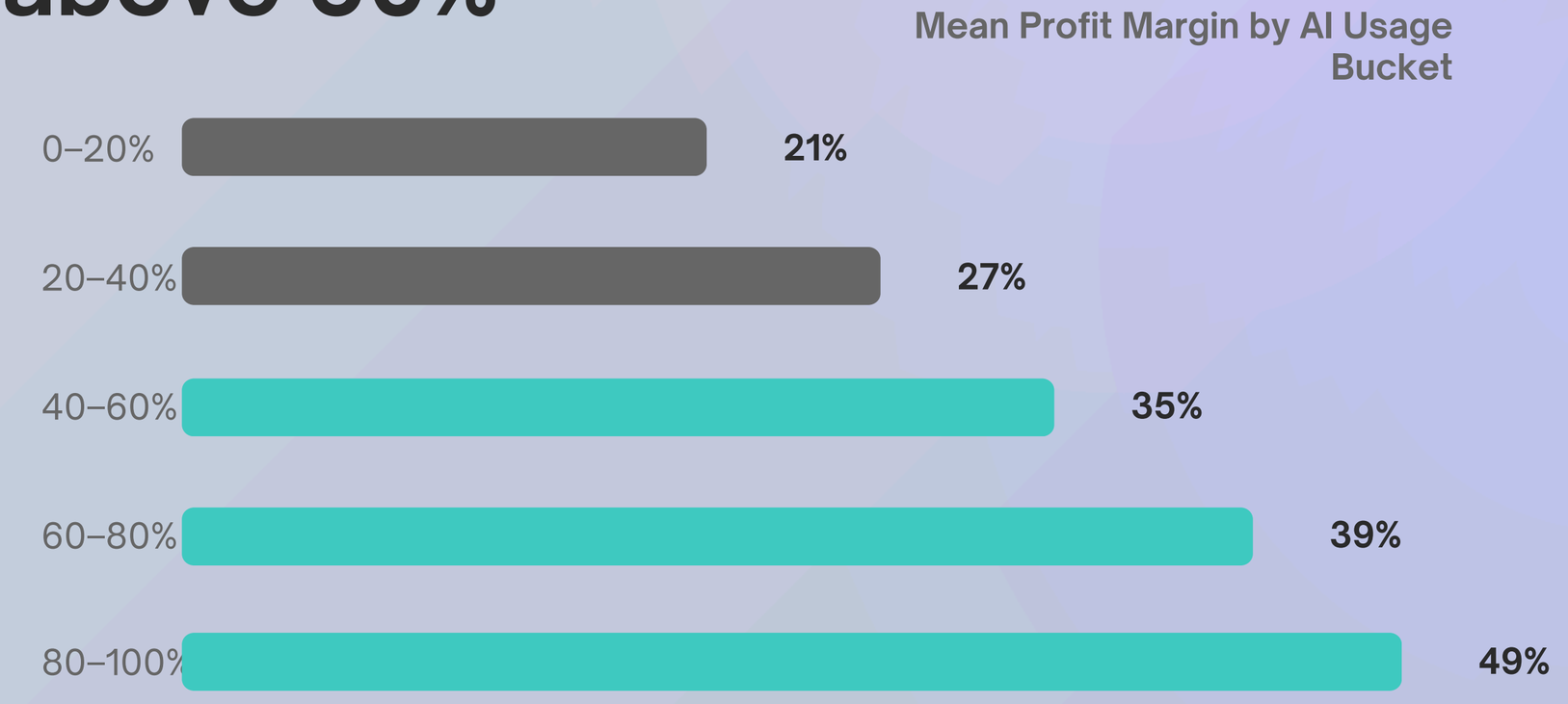
Is the damage done through **quality loss or** through **billable hours?**

Does the pricing structure determine who captures the **efficiency gain**?

[Alkemy]
Designed by humans. Powered by AI.

LUISS

AI improves margin: but the gains only consolidate above 50%



+28pp
margin gain from lowest to highest AI intensity bucket

−40%
SLA breach rate across the same range

45% → 27%
SLA breach

40–60%

Below this, rework absorbs the speed gain.

Above it, the efficiency dividend becomes real and measurable.

[Alkemy]
Designed by humans. Powered by AI.

LUISS

AI saves hours.
The contract decides who keeps them.

Rework cost shows near-zero correlation with profit margin (Spearman ρ = 0.02). The productivity gain from speed outweighs the correction cost: hours saved, not quality lost, is the real lever

HOW AI ACTS ON PROFIT

The productivity gain is real.

Hours spent falls from 11.6 hrs (0–20% AI) to 10.2 hrs (80–100% AI).

Rework rate doubles over the same range (0.133 → 0.298) — but the net time position stays positive.

CONTRACT TYPE DECIDES OUTCOME

Same time saved. Different P&L impact.

↑ Fixed / Value-based: hours saved = margin captured

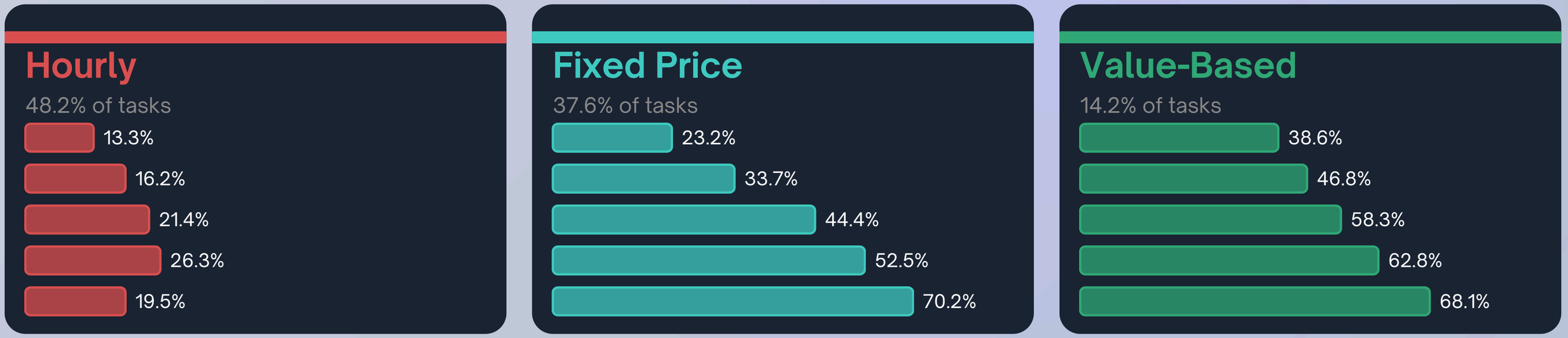
↓ Hourly billing: hours saved = revenue lost (peak at 26.3%, then falls to 19.5% at 80–100% AI)

[Alkemy]
Designed by humans. Powered by AI.

LUISS

Under hourly contracts, AI makes Alkemy faster for the client.
Not more profitable for itself.

Hourly margin peaks at 26.3% (60–80% AI), then falls to 19.5% at maximum intensity.
Fixed and value-based margins continue rising to 70.2% and 68.1%



[Alkemy]
Designed by humans. Powered by AI.

LUISS

Task type determines how far you can push AI.

Note: 80–100% bucket contains 48 tasks total. Terminal values carry high estimation uncertainty

Task Type	Range (0-20% → 80-100%)	Signal	Recommendation
Ticket	(11.3% → 58.7%)	+5.2×	Push AI hard: biggest multiplier in the dataset
Development	(19.8% → 70.8%)	+3.6×	Consistent gains: push to high intensity
Report	(19.3% → 60.2%)	+3.1×	Consistent gains: works well at high intensity
Design	(22.2% → 56.9%)	+2.6×	Good returns: push above 50%
Release	(16.3% → 39.6%)	+2.4×	Peaks at 60–80% (50.5%), slight drop after: cap at 80%
Article	(25.4% → 46.4%)	+1.8×	Medium band optimal: cap at 40–60%
Ad	(23.0% → 25.2%)	+1.1×	Minimal AI benefit: use AI selectively

[Alkemy]
Designed by humans. Powered by AI.

LUISS

THE THRESHOLD

The transition zone: where AI usage starts paying off

40–60%

Largest marginal gains occur crossing into 40–60% and at 80–100% intensity

Below 40%
AI Margin:
20.98% (median)

Largest marginal jump:

+8.76 pp crossing into 40–60% bucket.

Second peak:

+9.62 pp at 80–100%

Above 60% AI Margin: 39.34–48.96% (median)

[Alkemy]
Designed by humans. Powered by AI.

LUISS

What actually drives profit and where AI sits in that picture (SHAP Analysis)

#1

PRICING MODEL — THE STRUCTURAL DRIVER

pricing_model_hourly ranks first in SHAP. Its impact on predicted profit is larger than ai_usage_pct across every segment of the data.

#2

BILLABLE RATIO — THE EFFICIENCY SIGNAL

How much of worked time becomes billed time is the second-strongest predictor. It captures whether the team runs lean or absorbs hidden slack.

#3

AI USAGE: Rank 14

ai_usage_pct has a real but indirect effect. It does not independently move profit — it amplifies whatever structure already exists around it.

1. pricing_model_hourly 15.3 2. billable_ratio 12.5 3. seniority_senior 10.1 4. hours_spent 7.2 --> 14. ai_usage_pct 0.7 ← not a direct driver

[Alkemy]
Designed by humans. Powered by AI.

LUISS

MODEL VALIDATION

Three models, the same story

Model	Test R^2	MAE (pp)	CV R^2	Key insight
Linear Regression	0.205	32.1	0.083	Interpretable baseline
Lasso (LassoCV $\alpha = 0.57$)	0.205	—	—	Used as feature selector, not final model
Random Forest (GridSearchCV)	**0.345**	27.6	0.184	Best predictive model

$\Delta R^2 = +0.14$

RF vs LR — strong non-linearity confirmed. Threshold effects are real, not artefacts.

65% of variance unexplained

Missing: client pricing power, AI tool quality, team capacity at the moment of assignment.

[Alkemy]
Designed by humans. Powered by AI.

LUISS

KEY INSIGHT

Three things that data taught us

01 AI is a multiplier, not a solution

AI usage does not independently move profit margin:

it amplifies whatever pricing structure already exists.

It changes between fixed and hourly contract.

The contract is the variable, not the AI.

02 The binary view was wrong

Tasks with any AI usage showed lower median margin than tasks with no AI.

This seems damaging: until you treat AI as intensity rather than a flag.

At 80–100% AI usage, median margin reaches 49%.

The binary comparison was obscuring the real signal entirely.

03 Rework rises with AI — and so does profitability

We expected rework to be the margin killer.

Instead, rework rate doubles as AI intensity rises (0.133 → 0.298) while margin rises from 21% to 49% over the same range.

The paradox forced us to rethink the mechanism: the intervention point is pricing, not quality control.

[Alkemy]
Designed by humans. Powered by AI.

LUISS

What we found, what we built, and what went wrong

Concrete Decision

The constraint is not the technology: it is the contract

AI can improve profitability, but only when adoption is deep enough and the commercial structure is aligned to capture the gains. Getting one without the other leaves value on the table. Nearly half the dataset runs on hourly contracts. That is the structural constraint. Reprice first, then scale.

Thing discovered thanks to AI

The rework paradox: a finding that would have been dismissed as noise

Combining visual and tabular analysis tools revealed a counterintuitive trend: higher AI intensity increases both rework costs and profit margins at the same time.
Rework is actually an indicator of task complexity, rather than just an isolated expense.
Without looking through these methods, this paradox would have likely been ignored instead of being understood.

Mistake made by AI

AI kept pushing us to cap the outliers. We did not. That was the right call.

Standard data cleanup methods would have removed valid extreme values by treating them as errors.
Accepting the AI's suggestion would have skewed the results toward the average, hiding important edge cases.
AI offered a theoretically solution but lacked business context. Catching the mistake required human judgment to understand what the outliers actually meant, rather than treating data preprocessing as a mindless checklist.

[Alkemy]
Designed by humans. Powered by AI.

LUISS

Three decisions that follow directly from the data

1
Primary lever

Reprice before scaling AI further

Increasing AI adoption under hourly contracts transfers the efficiency gain to the client, not the business. The pricing book needs to change first.

2
Immediate

Set a 30% floor, target 60%

Below 30%, AI usage is on average a net cost. The transition zone is 30–60%. Above 60%, profit and time savings consolidate. Treat these as operational thresholds, not guidelines.

3
Highest ROI

Rework reduction beats AI restriction

68% of AI's margin effect flows through rework. Improving QA and brief quality generates more margin uplift than capping AI usage rates.

[Alkemy]
Designed by humans. Powered by AI.

LUISS

BLIND SPOTS

What the data cannot tell us: yet.

The model is honest about its limits. These are the variables that would change the recommendations.

01

CLIENT RETENTION

We measure per-task profit. We do not know whether AI-heavy delivery affects renewal rates, NPS, or long-term contract value.

HIGHEST PRIORITY

02

TOOL QUALITY

ai_usage_pct tells us how much AI was used — not how well. Two tasks at 80% can have completely different quality profiles.

PROCUREMENT

03

REVISION CAUSALITY

Rework hours capture cost, not cause. We cannot distinguish AI-generated errors from scope changes or client preference shifts.

ROOT CAUSE

[Alkemy]
Designed by humans. Powered by AI.

LUISS

The AI is working. The pricing model isn't.

RESTRUCTURE THE CONTRACT BEFORE YOU SCALE THE TECHNOLOGY.

Group SEA — Simone Cavallaro · Andrea Mammano · Elisa Presciutti · LUISS 2025/2026